BrightLearn Data Analytics

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Exercise 4: UNION & UNION ALL

Question 1-3 (UNION)

(1)
```
(SELECT account_id,
        account_holder
 FROM branch_sandton_accounts)
 UNION
(SELECT account_id,
        account_holder
 FROM branch_rosebank_accounts);
```

account_id	account_holder
A001	Nomvula Dlamini
A002	David Mokoena
A003	Lerato Sithole
A004	Peter Nkosi
A005	Zanele Khumalo
A006	Thabo Motha

(2)
```
(SELECT product_code,
        product_name
 FROM saving_products)
 UNION
(SELECT product_code,
        product_name
 FROM current_products);
```

product_code	product_name
SV01	Basic Savings
SV02	Premium Savings
SV03	Youth Savings
SV04	Business Saving
CR01	Standard Current
CR02	Gold Current
CR03	Business Current

3) (SELECT staff_id,
staff_name
~~UNION~~ FROM retail_banking_staff)
UNION
(SELECT staff_id,
staff_name
FROM corporate_banking_staff);

staff_id	staff_name
SC01	Mpho Radebe
SC02	Brian Tshabalala
SC03	Aisha Patel
SC04	Kabelo Moabelo
SC05	Nandi Dube
SC06	Sipho Khumalo

4) (SELECT city_code,
city_name
FROM mobile_branch_cities)
UNION
(SELECT city_code,
city_name
FROM digital_branch_cities);

city_code	city_name
c01	Johannesburg
c02	Pretoria
c03	Cape town
c04	Durban
c05	Polokwane
c06	Port Elizabeth

5) (SELECT customer_id,
customer_name
FROM push_notification_targets)
UNION
(SELECT customer_id,
customer_name
FROM inapp_banner_targets);

customer_id	customer_name
C1001	Nomsa Zwane
C1002	Andile Ruthelezi
C1003	Fatima Mahomed
C1004	Ryno van Zyl
C1005	Thandeka Cele
C1006	Samuel Nkosi

Questions 6-10 (UNION ALL)

(6)
```
(SELECT txn_id,
        account_id,
        amount
 FROM atmo1_transactions)
UNION ALL
(SELECT txn_id,
        account_id,
        amount
 FROM atmo2_transactions);
```

txn_id	account_id	amount
T1001	A001	500.00
T1002	A002	1200.00
T1003	A001	800.00
T1004	A003	750.00
T1003	A001	800.00
T1005	A004	900.00
T1006	A002	450.00
T1007	A005	150.00

(7)
```
(SELECT app_id,
        customer_id,
        loan_type
 FROM gauteng_loan_applications)
UNION ALL
(SELECT app_id,
        customer_id,
        loan_type
 FROM western_cape_loan_applications);
```

app_id	customer_id	loan_type
LA001	c1001	Home Loan
LA002	c1002	Vehicle Loan
LA003	c1003	Personal
LA004	c1004	Home Loan
LA003	c1003	Personal
LA005	c1005	Vehicle Loan
LA006	c1006	Business
LA007	c1007	Home Loan.

(8)
```
(SELECT complaint_id,
        customer_id
 FROM email_complaints)
 UNION ALL
 (SELECT complaint_id,
         customer_id
 FROM app_complaints);
```

complaint_id	customer_id
Ec001	c2001
Ec002	c2002
Ec003	c2003
Ec004	c2004
Ac001	c2005
Ac002	c2001
Ac003	c2006
Ac004	c2007

(9)
```
(SELECT payment_id,
        account_id,
        amount
 FROM april_payments)
 UNION ALL
 (SELECT payment_id,
         account_id,
         amount
  FROM may_payments);
```

payment_id	account_id	amount
PAY001	A001	12500.00
PAY002	A002	4500.00
PAY003	A003	9200.00
PAY004	A004	3300.00
PAY005	A001	12500.00
PAY006	A005	7600.00
PAY007	A002	5100.00
PAY008	A006	2800.00

(10)
```
(SELECT entry_id,
        account_id,
        entry_type
 FROM debit_entries)
 UNION ALL
 (SELECT entry_id,
         account_id,
         entry_type
  FROM credit_entries);
```

entry_id	account_id	entry_type
DR001	A004	Debit
DR002	A002	Debit
DR003	A003	Debit
DR004	A004	Debit
CR001	A001	Credit
CR002	A005	Credit
CR003	A002	Credit
CR004	A006	Credit

(03) Bonus Questions

Bonus 01

A fraud analyst should use UNION

Explanation : It automatically removes duplicates, ensuring each flagged customer appears only once.

Bonus 02

The auditing team should use UNION ALL

Explanation : It retains all rows without deduplication, preserving the total count needed for the audit.

Bonus 03

The statement returns an unequal number of columns which breaks the column-count rule.

To fix this we match the columns in both SELECT statements.